

BMET3997/9997: Major Project

QRS detection and heart rate variability measurement from a single lead ECG.

Semester 1, 2026 | 6 Credit Points | Coordinator(s): Philip de Chazal
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Over the next six weeks you will work in groups 3 students) to develop a system that automatically identifies QRS complexes and estimates heart rate variability parameters from single lead electrocardiographic (ECG) signals.

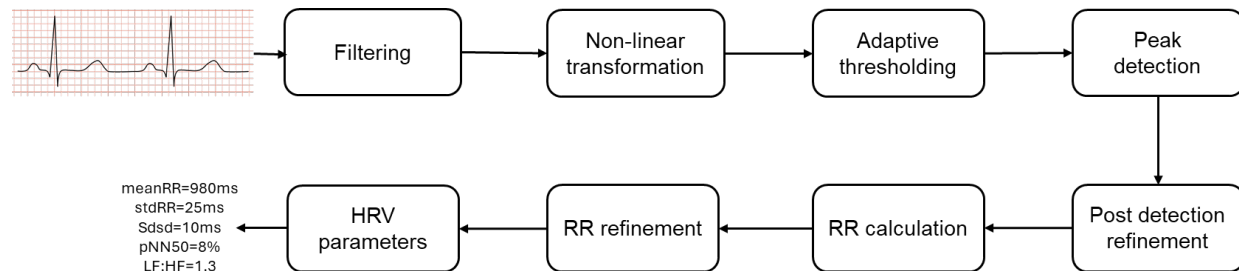


Figure 1: Outline of a possible sequence of signal processing steps.

1. Each group will work to achieve the following

- Use provided training data, develop a QRS detection algorithm and HRV calculation algorithms that will process the provided training and test data. The format of test data is the same as the training data
- Submit to lab tutor (Siying) via email, your QRS detection times and your estimated HRV parameters for the test data. We will then assess scores against the test data and calculate the performance metrics and inform you of the performance.
- Each group will get a maximum of 5 submissions sent to the lab. tutor between weeks 7 and week 13.
- The best set of performance metrics for each group for the test data will be displayed on Canvas and each group will compete to get the highest score.

2. Training data

- Available on Canvas under the Major Project Module **ProjectTrainData.mat**
- The .mat file contains two cell arrays
 - **ECG**: a cell array with 35 overnight ECG each sampled at 100Hz and signal amplitude in uV. ECG{1} contains the ECG signal for overnight recording 1, ECG{2} contains the ECG signal for overnight recording 2, etc. Note that each night has a different length.
 - **QRSexpert**: a cell array with the expert validated QRS points for each of the ECG recordings. QRSexpert{1} contains the QRS detection points (in samples) for overnight recording 1. QRSexpert {2} contains the QRS detection points for overnight recording 2, etc. Note that each night has a different length.

3. Testing data

- Available on Canvas from week 9 onwards under the Major Project Module **ProjectTestData.mat**
 - The ProjectTestData.mat file contains 35 overnight ECG recordings sampled at 100Hz and signals amplitude in uV.
- The ProjectTestDataAnalysis.mat contains a cell array for the QRS detections and arrays for the HRV parameters.

4. ProjectTestDataAnalysis.mat

This file has two groups of variables: QRS and HRV parameters

QRS

- As they currently are
 - $\text{QRS}\{1\}' = [100, 200, 300]$
- Once ready for a submission, you will need to change the variables to something like
 - $\text{QRS}\{1\}' = [56, 148, 276, 512, \dots]$
- I expect there will be between 20-50k detections for each of the 35 recordings.

HRV parameters

You will divide the ECG into non-overlapping 5-minute windows (window 1 will cover 0-5 minutes, window 2 will cover 5-10 minutes etc) and calculate the following HRV parameters for RR intervals in each window. The units are shown in [].

- The average RR interval (avgRR [ms])
- The standard deviation of the RR intervals (sdRR [ms])

- The square root of the mean of the squares of differences between adjacent RR-intervals (RMSSD [ms])
- Percentage of successive RR interval that differ by more than 50 ms (pNN50[%])
- The low frequency power (LF [ms²])
- The high frequency power (HF [ms²])
- The low frequency to high frequency power ratio (LF:HF [no units])

Once you have the HRV parameters for each window, you will then average them across the windows so you have one value for each of the HRV parameter for each record.

Note for some windows, you may decide to not calculate HRV parameters for reasons including signal too noisy or too many unreliable QRS detections. When you average across the windows, you will need to ignore the HRV parameters from these windows.

There is one array for each HRV parameter in ProjectTestDataAnalysis.mat and they are currently set to NaN. You will replace these values with your averaged values from above.

For example, for record 1

- As they currently are
 - avgRR[1]=nan; sdRR[1]=nan; RMSSD[1]=nan; pNN50[1]=nan; LF[1]=nan; HF[1]=nan; LF_HFratio[1]=nan;
- Once ready for a submission, you will need to change the variables to something like
 - avgRR[1]=950; sdRR[1]=45; RMSSD[1]=34; pNN50[1]=12; LF[1]=605; HF[1]=495; LF_HFratio[1]=1.2;

Submission

- Save your updated values to a new .mat file. Please DO NOT include the ECG variables from ProjectTestData.mat in your submission. The new file name will contain your group number and submission number e.g. ProjectTestDataAnalysisGroup2Submission3.mat
- You will need to make sure that you do not transpose the variable QRS.
- You will submit your updated ProjectTestDataAnalysisGroup*Submission*.mat to lab tutor (Siyong) as a submission.
- We will use it to calculate a) sensitivity, positive predictivity and F1-score of your QRS detection points and b) mean absolute percentage error of your HRV parameters using the expert-based QRS detection points (which we will keep secret) and report back to you.
- Towards end of the semester, I may also get you to submit your training set QRS detections and HRV parameters of your best system so we can assess the training set performance of your system.

5. Performance Assessment of your QRS detections

- Performance will be assessed using Sensitivity, Positive Predictivity and F1-score from the TP, FP and FN counts across the test recordings
 - A TP will be counted when your QRS detection is within or equal to 50ms of a true QRS detection.
 - A FP will be counted when your QRS detection is more than 50ms from all true QRS detections.
 - A FN will counted when a true QRS detection is more than 50ms from all your QRS detections.

6. Performance Assessment of your HRV parameters

- Each HRV parameter provided by you will be assessed using the mean absolute percentage error (*MAPE*) calculated using the reference set of HRV parameters from the 35 test set recordings. An example is shown for *avgRR*
 - $$MAPE_{avgRR} = \frac{100}{35} \sum_{i=1}^{35} \left| \frac{avgRR_{ref}(i) - avgRR_{yours}(i)}{avgRR_{ref}(i)} \right|,$$
where *avgRR_{ref}* is the reference value and *avgRR_{yours}* is your value.
- The reference set of HRV parameters was calculated as follows. First, a reference set of RR intervals was calculated from the RR intervals determined from the true QRS detections. Any RR intervals that were physiologically unlikely have been flagged (i.e RR interval were flagged if $RR > 2000\text{ms}$ or $RR < 250\text{ms}$). Finally, HRV parameters were calculated for each non-overlapping 5-minute window based on the unflagged RR intervals and then averaged across the recording, resulting in the reference set of HRV parameters. Only windows with a minimum of 4 minutes of accumulated unflagged RR intervals are included in the average.

7. Key points

You are free to use any of the toolboxes in Matlab and libraries in Python except for toolboxes or libraries that relate to QRS detection or HRV calculation. If you are unsure ask Siying of myself.

- a. You must provide a list of all toolboxes and libraries in your final report.
- b. Python implementations must be able to process the *.mat file as input and produce the required *.mat as output.
- c. We will provide code that will help with this conversion

QRS detection

- All students: You may design a single or multi-pass algorithm QRS detection
- BMET3997 students:
 - You may process the ECG in forwards and reverse time
- BMET9997 students:

- You may process the ECG in forwards time only so that your final algorithm could potentially be used in a real-time application.

HRV calculation

- As your algorithm will rely on your calculated QRS detections from above, you should design your calculation methods so that they are robust to imperfect or missing QRS detections.

8. Submissions

- Each group will get a maximum of 5 submissions sent to the lab. tutor between weeks 7 and Friday 5pm week 13.
- If your submission is in the wrong format then we will let you know and you can submit again without using up one of your submissions.
 - It is up to your group to make sure you have the correct format!

9. Summary

- Major areas of technical work to be finished by week 13 include
 - Implement a QRS detector as a first stage
 - Implement a second stage algorithm that assesses QRS detection points from the first stage and attempts to correct possible false positives and false negatives (post-detection refinement).
 - Implement algorithms for calculation of required HRV parameters in the presence of imperfect or missed QRS detections.
- Non-technical work
 - Final report (Tuesday 1st week exam period)
 - Presentation (Monday week 13)

Formats for the report and presentation will be provided later in the semester

10. Preliminary signal processing exercises

Load the data

- Available on Canvas under the Major Project Module **ProjectTrainData.mat**

Load ApnoeaECG.mat into Matlab using `load ProjectTrainData.mat`

There are 2 cell arrays. Each cell contains the data for 1 record

- ECG. This contains the ECGs sampled at 100Hz and 1 bit = 0.005mV. Each cell contains the ECG signal for 1 record e.g. `ECG{10}` contains the ECG signal of the 10th recording
- QRSexpert. This contains the QRS detection points in samples. Each cell contains the QRS signal for 1 record e.g. `QRSexpert{35}` contains the QRS signal of the 35th recording

Try the following commands to get a feel for the data

- 1) Display the first one hundred samples of the ECG in the 1st record by typing
`ECG{1}(1:100)`
- 2) `QRSexpert{1}` contains the QRS detection points (in samples) for overnight recording 1. The first 3 values are 40, 128 and 213 which tells you that there is a QRS peak at 40, 128 and 213 samples. The following Matlab code will visually confirm this for you

```
>> plot(ECG{1}(1:300));  
hold on;  
plot(QRSexpert{1}(1:3), ECG{1}(QRSexpert{1}(1:3)), 'go');  
xlabel('Samples')  
ylabel('uV')
```

- 3) Display the length of the first ECG in hours (8.21 hours)

```
>> length(ECG{1}) / (100*3600)  
ans =
```

```
8.2139
```

Run a QRS detector algorithm over the first ECG recording

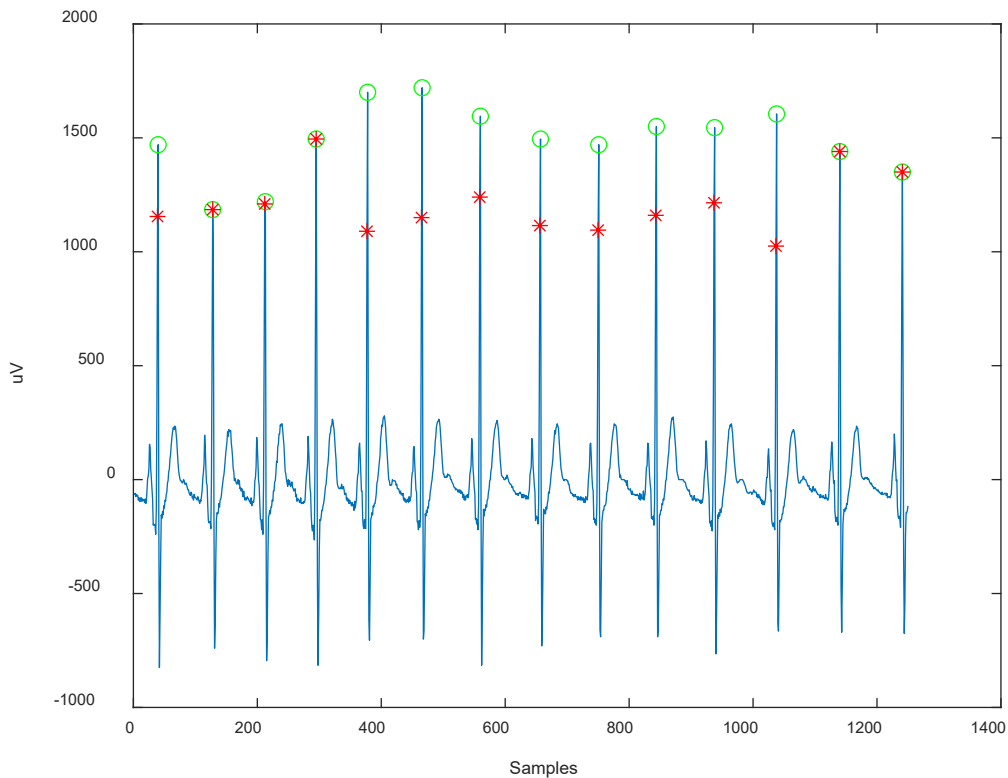
We will now process the first ECG with a Pan-Tompkins QRS detector. This file has been obtained from Matlab File Exchange. First, download the file `pan_tompkin.m` from Canvas.

```
>> fs=100;  
gr=false;  
[~,QRS{1}]=pan_tompkin(ECG{1},fs,gr);
```

We can now replot the first ECG and show both the expert determined QRS points (green circles) and the pan-tompkins determined QRS points (red asterisk)

```
>> clf  
plot(ECG{1}(1:1250));  
hold on;  
plot(QRSexpert{1}(1:14),ECG{1}(QRSexpert{1}(1:14)),'go');  
plot(QRS{1}(1:14),ECG{1}(QRS{1}(1:14)),'r*');  
xlabel('Samples')  
ylabel('uV')
```

Your plot should look like



Estimate performance of Pan-Tompkins algorithm on first record

We estimate the performance of this QRS detector by first counting the true positives (TP), false positives (FP) and false negatives (FN), and the calculating our performance figures. Note the following:

- A TP if a QRS detection is within 50ms of an expert determined QRS detection.
- A FN if an expert QRS detection is more than 50ms from all QRS detections.
- A FP if a QRS detection is more than 50ms from all expert determined QRS detections.

```
>> % Calculate TP, FP and FN counts across this recording
Tol=5; %+/-50ms
TP=0;
FP=0;
FN=0;

for j=1:length(QRSexpert{1})
    if any(abs(QRSexpert{1}(j)-QRS{1})<=Tol)
        TP=TP+1;
    else
        FN=FN+1;
    end
end
for j=1:length(QRS{1})
    if all(abs(QRS{1}(j)-QRSexpert{1})>Tol)
        FP=FP+1;
    end
end
```

The next step is to calculate sensitivity, positive predictive value and F1-score. These are defined as follows

- $Sens = \frac{TP}{TP+FN}$
- $PPV = \frac{TP}{TP+FP}$
- $F1 = 2 \times \frac{Sens \times PPV}{Sens+PPV}$

Using Matlab....

```
>> Sens=TP/(TP+FN)
    PPV=TP/(TP+FP)
    F1=2*Sens*PPV/(Sens+PPV)
```

This should result in the following values.

Sens = 0.9999; PPV = 0.9996; F1 = 0.9997

Which suggests the QRS detector works very well for this record!

Estimate performance of Pan-Tompkins algorithm on all the training data

Your turn! Calculate the sensitivity, positive predictivity and F1-score after accumulating the TP, FP and FN over the entire set of 35 recordings. You should get the following figures.

Sens = 0.7818
PPV = 0.9863
F1 = 0.8722

This performance is much worse than what was obtained for ECG 1. Investigate the performance on individual ECGs and determine which records the performance is not so good.

Calculate the avgNN HRV parameter from the Pan-Tompkins QRS detections record 1 for the first 5 minute window

We first calculate the RR values from the QRS values using the diff function. Note there is always one less RR interval the QRS detections points. For future processing we need the two sequences the same length, so we chop off the last QRS point.

Matlab code to achieve this is as follows.

```
for i=1:35
    RR{i}=diff(QRS{i})/100; %RR interval in seconds
    QRS{i}(end)=[]; %Make sure two sequences are the same length
end
```

One of the easiest heart rate variability parameters to calculate is the average heart rate in a window. The following code will find this value for the first 5 minute window.

```
RRseg=RR{1}(QRS{1}>=1 & QRS{1}<=5*100*60);
avgRRepoch=mean(RRseg)
```

and results in

avgRRepoch = 0.8159

for the window

Calculate the avgRR HRV parameter from the Pan-Tompkins QRS detections for record 1 for all 5 minute windows and then average

Your turn! Extend the code above and calculate the average value over all 5 minute windows in record 1

You should get the following value

```
avgRR = mean(avgRRepoch);
avgRR = 0.9921
```

Now extend this code over all 35 records

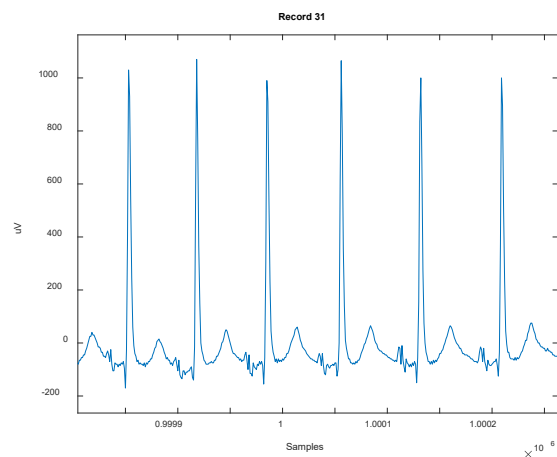
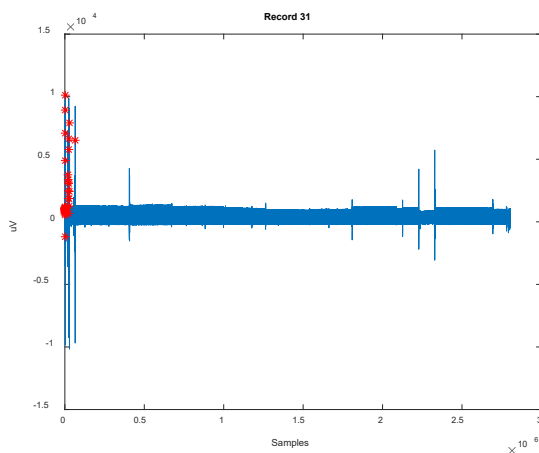
This should result in the following values

```
avgRR =
[ 0.9921    0.7464    0.8875    0.9599    0.9473    0.9860    0.8201
  0.7309    0.9470    0.9622    0.8480    0.9646    0.7478    1.0857
  0.9042    0.8303    1.0438    0.9776    0.7772    0.8879    0.8352
  0.9151    0.9103    3.5125    56.2500    2.9033    12.7803    1.1491
  1.6009   7546.1    1132.5    0.8053    0.9865    1140.1    3004.2]
```

You will notice that some of the average RR values for the last 15 records are clearly wrong as they are 3-7500s.!

To investigate this try plotting the ECG for record 31 and overlaying the QRS detections by using

```
>> plot(ECG{31});
    hold on;
    plot(QRS{31},ECG{31}(QRS{31}),'r*');
    xlabel('Samples')
    ylabel('uV')
```



The Pan-Tompkins QRS detector has completely failed on this record even though there are good quality ECG signals.

Calculate MAPE for avgRR parameter

The final step is to calculate the MAPE for the avgRR parameter. The reference set of avgRR values for the 35 training records are shown below. These have been calculated from the expert determined QRS detections and have been corrected by 1) removing all RR intervals greater than 2 seconds and all RR-intervals less than 0.25 seconds and 2) only including 5 minutes intervals with a minimum of 4 minutes of valid RR intervals. This was previously described in section 6 above.

```
avgRRexpert =  
    0.9924    0.7498    0.9218    0.9599    0.9487    0.9904    0.8195  
    0.7310    0.9472    0.9631    0.8496    0.9765    0.7480    1.0855  
    0.9053    0.8328    0.7880    0.9782    0.7762    0.8888    0.8359  
    0.9018    0.9129    1.0324    0.9430    1.0307    0.9221    1.1506  
    0.9900    0.9991    1.0009    0.8094    1.0217    0.8799    1.0535]
```

The MAPE value for avgRR using the Pan-Tompkins detector is

```
MAPE=mean(abs((avgRR-avgRRexpert)./avgRRexpert))*100;  
MAPE = 65962.
```

This is a very poor result as the average percent error is almost 66000%! Using this Pan-Tompkins detector, how can you improve this result?

The results are much better if limit our focus the first 20 records (which excludes all the clearly wrong records)

```
MAPE = 1.9830.
```

11. Complete the major project!

Having completed the above exercises, you can now start replacing/improving these algorithms for the major project.

12. HRV parameters values for expert annotations on training data

Recording	avgRR [ms]	sdRR [ms]	RMSSD [ms]	pNN50 [%]	LF [ms ²]	HF [ms ²]	LF_HFratio [no units]
1	992	143.8	77.1	36.4	8947	1910	5.1
2	750	45.3	22.4	1.6	807	109	15.08
3	922	116	77.5	15.7	2446	959	2.88
4	960	84.6	42	9	1913	414	7.55
5	949	71.4	31.8	8.4	1268	375	4.02
6	990	63.4	52	18.7	1200	977	1.59
7	820	102.9	59.5	16.8	5207	1275	4.24
8	731	47.9	25.9	2.3	716	139	6.04
9	947	52.1	40.4	1.5	473	239	3.77
10	963	63.3	34.3	8.1	1088	509	2.64
11	850	43.1	31.4	6.2	612	363	2.33
12	977	94.5	42.8	11.6	2754	455	9.1
13	748	56.9	30.7	3	1089	161	7.62
14	1086	141.4	90.4	44.5	8361	2919	3.15
15	905	78.6	28.3	3.5	2267	230	16.58
16	833	78.3	37.5	9.2	2427	495	5.72
17	788	77.1	68.6	16.1	1758	2097	2.74
18	978	44.3	34.5	1.1	511	199	4.96
19	776	33.7	13.9	0.2	350	64	8.63
20	889	52.4	24.3	2.7	847	230	4.76
21	836	64.3	35.7	10.4	1525	558	3.53
22	902	50.9	20.1	1.1	894	127	11.07
23	913	48.6	20.9	1	748	138	7.88
24	1032	87	53.8	22.8	3402	911	4.23
25	943	110.9	152.2	39.1	1258	4133	0.39
26	1031	87.9	66.8	32.7	3356	1702	2.37
27	922	39.5	23.9	2.9	394	184	2.58
28	1151	145	142.8	56.6	8379	6441	1.43
29	990	206.6	328.2	71.5	1412	17240	0.2
30	999	69.6	53.7	22	1753	1257	1.99
31	1001	70.7	54	22.6	1793	1174	2.06
32	809	52.8	29.8	3.5	1020	423	3.94
33	1022	82.6	53.7	24.7	3233	939	4.27
34	880	39.3	22	2.6	460	172	3.27
35	1054	74.1	46.7	19.7	2173	979	2.16